

Schedule 1-B

SCOPE OF WORK - UpSCALedEV

This scope of work document provides details about the project objectives, work package breakdown, deliverables and timeline of the UpSCALedEV project.

Project objectives:

The overall objective of this proposed research work is to develop and test coordination and dispatch strategies for behind-the-meter distributed energy resources (DERs) to reduce commercial and industrial (C&I) customer utility bills and/or greenhouse gas (GHG) emissions. The focus is on dispatching Electric Vehicles leveraging optimization strategies and IoT hardware developed under the SCALedEV, coDERMS, and DERConnect projects, to provide a solution with quantifiable customer benefits.

Three specific sub-objectives are:

1. **Set up interconnected DER coordination and dispatch platform:** Choose a building load, battery energy storage system (BESS), and electric vehicle (EV) charging plaza that are physically or can be virtually interconnected.
2. **Develop optimization framework:** Develop the optimization framework to measure costs and benefits of DER dispatch. In particular, cost functions, application scenarios, and market operation workflows will be defined.
3. **Run simulated and real trials at one site:** Run simulated and real trials to measure the success of the dispatch strategies and provide recommendations for practical implementation.
4. **Upscaling:** Simulate upscaling of the results from a single site to a portfolio of sites and investigate the transferability of optimization framework to different markets in the US.

Benefits of the proposed UpSCALedEV project include:

1. A testbed that includes all assets commonly found at commercial and industrial (C&I) customers. Battery energy storage systems, electric vehicles, and building loads (Heating, Ventilation, and Air Conditioning (HVAC), plug loads, and lighting) are available for dispatch.
2. High-frequency (2 second) metering is in place to permit intra-15 minute dispatch adjustments for demand charge management.
3. Complex optimization algorithms that integrate market participation, demand charge management, low carbon fuel standard revenue, and time-of-use rates are available, and transferability to different markets (different market rules, policy and grid service requirements)
4. Potential use of the campus testbed as an indicator for nodal load impact when adding DERs including EV charging, PV and Storage

Work package breakdown and technological approach:

The work plan has been divided into four work packages (WP) as explained below.

WP 1: Set up interconnected DER coordination and dispatch platform

The focus of this WP is to set up an interconnected DER coordination and dispatch platform that will be leveraged in WP3 to run tests.

- 1.1. Catalogue available DERs including EV charging stations, energy storage systems at different sites. Buildings will be treated as uncontrollable loads. See Table 1 and Figure 1 for an example at the Hopkins Parking Structure. See Table 2 for all available campus DERs. Characterize power and energy capacities, communication protocols, availability, and constraints.

Table 1: DERs at the Hopkins Parking Structure

Type	Specs	Communications	Status
Photovoltaic	338 kW _{DC} , 350 kW _{AC}	IEEE 2030.5	Operational
Lithium-Ion Battery Energy Storage System (BESS)	250 kW, 558 kWh	IEEE 2030.5	Planned
Electric Vehicles	20x 6.6 kW PowerFlex L2 chargers	PowerFlex API	Operational
Building Load	466 MWh in 2023	N/A Static Load	Operational

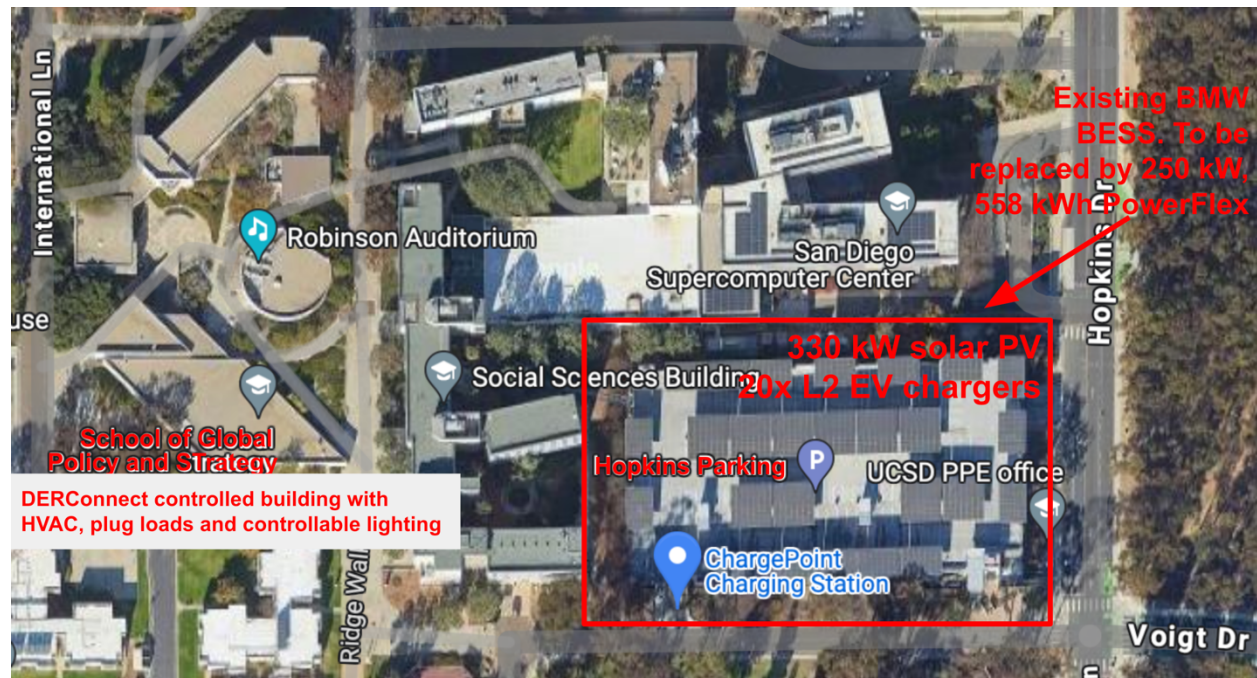


Figure 1: Map of the DERs at Hopkins Parking Structure and adjacent buildings

Table 2: Summary of controllable DERs on the UC San Diego Campus through DERConnect

Type	Specs	Communications	Status
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Photovoltaic	2 systems with IEEE 1547-2018 compatible inverters	IEEE 1547-2018	In design
Lithium-Ion BESS	250 kW, 558 kWh at Hopkins Parking Structure 10,000 kW, 40,000 kWh East Campus 250 kW, 332 kWh DERConnect 125 kW, 221 kWh DERConnect 2x 40 kW, 78 kWh DERConnect	IEEE 1547-2018	Funded Funded Under construction
Ultracapacitor	120 kW, 28 kWh DERConnect	IEEE 2030.5	U/C
Electric Vehicle chargers	600x 6.6 kW PowerFlex L2 chargers	PowerFlex API	300 operational, 300 U/C
Electric Vehicle Chargers	24x 62.5 kW ChargePoint DC Fast Chargers	ChargePoint API	Presently not controllable
Air handlers	30 AHU in 11 buildings with 1,170 thermostats and 237 submeters	BACnet	Operational
Connected LED lighting system	100 kW in 2,000 fixtures in 4 buildings	Acuity API	Operational
Plug loads	683 plug loads in 14 buildings	BACnet	Operational

1.2. Select a field site(s) for an interconnected testbed of DERs.

1.3. Configure DER Management System (DERMS) to communicate with all DERs and enable control trials (Figure 2). Commission system by conducting communication and dispatch tests with individual DERs.

WP 2: Develop Optimization Framework

The focus of this WP is to develop the optimization framework and implementation workflows to dispatch the DERs.

2.1. Cost functions: The following optimization cost function terms will be developed: peak demand charge, non-coincident demand charge, (volumetric) time-of-use, Low Carbon Fuel Standard (LCFS) revenue, wholesale market participation, carbon cost, event-based demand response (based on emergency load reduction program (ELRP)), frequency regulation. Update the cost functions for different US markets (CAISO, ERCOT, PJM).

2.2. Application scenarios:

- i. Regular C&I customer: demand charge, time-of-use rates, LCFS
- ii. Demand response C&I customer: (i) + demand response
- iii. Environmental C&I customer: (ii) + carbon cost
- iv. Sophisticated C&I customer: (iii) + wholesale market participation, frequency regulation

- 2.3. Workflows: Real-world implementation requires forecasts for load, generation, and prices. All prices will be assumed to be known (perfect forecast). DER availability, load, and generation will be forecast. The optimization workflow will consist of a day-ahead optimization that determines market participation, a real-time optimization that schedules DERs, and an intra-15 minute optimization that adjusts BESS and EVs to maintain target power levels.

Work Package 3: Simulated and Real Trials

This WP incorporates trials that apply the optimization framework from WP2 to the DER coordination and dispatch platform developed in WP1. Simulated trials will be run on one year of data. Real trials will be run for 20 working days. Results will be tabulated to compare cost performance under different scenarios.

- 3.1. Simulated trials: All scenarios and cost functions developed in WP2 will be applied to a 1-year dataset.
- 3.2. Real trials: Based on the results from the simulated trials two application scenarios will be selected for a real trial that will be executed on 20 business days. Real DERs that will be included are PowerFlex EVs (consented drivers from the Triton Charging Club), BESS, solar PV, and building loads.
- 3.3. Recommendations: Based on the outcomes of the simulated and real trials, recommendations will be developed for applying the research results to C&I customers.

Work Package 4: Upscaling

The goal of this WP is to upscale the findings from a single site to a portfolio of 10 sites. In the real world, virtual power plants are often created to aggregate individual sites for economies of scale. The effects of optimizing individual sites on the performance of the site portfolio will be investigated in simulations. Exclusions: A separate optimization for the portfolio is not included in the scope.

- 4.1. Site Selection: Select 10 sites with a mix of DERs. Obtain DER specifications for these sites as in WP1.1.
- 4.2. Portfolio simulations: The scenarios and cost functions of WP2 will be applied to the 1-year dataset in WP3.1 for the sites in WP4.1. The performance of the portfolio of sites will be quantified. Their impact on the local feeder and nodal load level will also be analyzed.
- 4.3. Transferability: Analyze transferability of the methodology to other markets including CAISO, ERCOT and PJM. Update the cost functions of WP2 to calculate demand charge savings and market participation revenue in other markets. Rerun the portfolio simulations of Task 4.1.

Background Figures

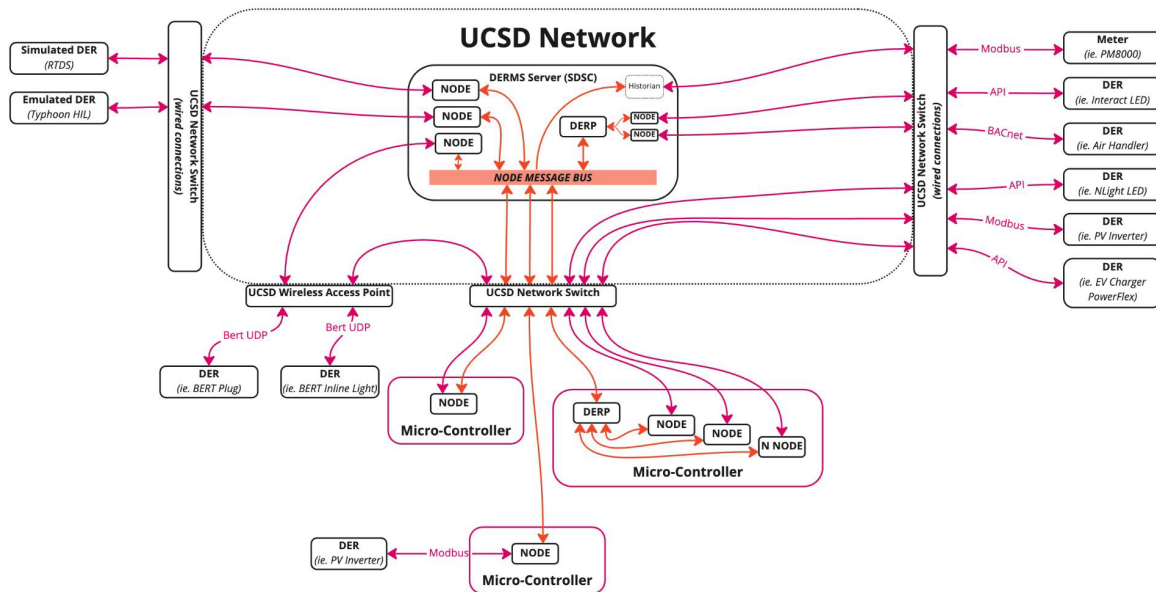


Figure 2: DERConnect communication framework.

Deliverables:

WP 1: Set up interconnected DER coordination and dispatch platform		Deliverables
1.1	Catalog available DERs	Table of available DERs at 10 sites including power and energy capacities, communication protocols, availability, and constraints.
1.2	Select field site	Field site name and description of characteristics.
1.3	Configure DERMS	Checklist of successful communications and dispatch tests.

WP2: Develop optimization framework		Deliverables
2.1	Cost functions	List of equations and parameter definitions of all cost functions.
2.2	Application scenarios	List of definitions of all application scenarios.
2.3	Workflows	Flowchart describing the temporal sequence, inputs, outputs, and cost functions of all optimization steps. Source code and documentation for the developed optimization algorithms.

WP3: Simulated and Real Trials		Deliverables
3.1	Simulated trials	Table showing the annual outputs of all cost functions for all optimization scenarios.

		Source code and documentation for the simulations.
3.2	Real trials	Table showing the daily values of all cost function terms. Slide show showing daily figures with dispatch results for all DERs.
3.3	Recommendations	Report describing recommendations for applying the research results to C&I customers.

WP4: Upscaling		Deliverables
4.1	Site selection	Ten field site names and description of characteristics.
4.2	Portfolio simulations	Table showing the annual outputs of all cost functions for all optimization scenarios. Report on nodal load impact. Source code and documentation for the simulations.
4.3	Transferability	Table showing the annual outputs of all cost functions for all optimization scenarios for PJM and ERCOT markets. Source code and documentation for the simulations.

Reporting:

Four reports will be delivered, one for each WP. The schedule for each report is provided below.

Report	Due date
WP 1	05/31/2025
WP 2	09/30/2025
WP 3	03/31/2026
WP 4	06/30/2026

Workshops/Tutorials:

Two workshops will be organized for knowledge transfer, tentatively in 2025Q3 and 2026Q2.

Timeline:

The anticipated project start date is 11/01/2024 and the duration of the project is 20 months.

WP schedule:

Work Package	Start	Duration (months)	Year 1				Year 2		
			24Q4	25Q1	25Q2	25Q3	25Q4	26Q1	26Q2
1.1	t_0	1	X						
1.2	$t_0 + 1$	1	X						
1.3	$t_0 + 5$	2			X				
2.1	t_0	2	X						
2.2	$t_0 + 2$	1		X					
2.3	$t_0 + 3$	2		X					
3.1	$t_0 + 8$	6				X	X		

3.2	$t_0 + 9$	8				X	X	X	
3.3	$t_0 + 13$	2					X	X	
4.1	$t_0 + 9$	1				X			
4.2	$t_0 + 11$	3					X		
4.3	$t_0 + 14$	6						X	X

Schedule 2-B

See attached excel file for detailed budget items